

Open-Source Licensing & IP Compliance Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author / owner of proprietary code	Samuele Pio Provenzano

1. Third-party dependency matrix

All bundled third-party libraries are **open-source under permissive licenses** (MIT / BSD-3-Clause). **No viral copyleft dependency (GPL / AGPL / LGPL)** is used, so there are no obligations to redistribute the proprietary code.

Component	Repo file	Use	License
RDKit (MinimalLib / rdkit-js)	RDKit_minimal.js , RDKit_minimal.wasm	Cheminformatics engine (SMILES/InChI/InChIKey/SMARTS, descriptors, fingerprints, 2D depiction)	BSD-3-Clause
3Dmol.js	3Dmol-min.js	3D viewer for molecules/macromolecules (WebGL)	BSD-3-Clause
three.js	three.min.js , three_bloom.js , glTF_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms)	MIT
SmilesDrawer	smiles-drawer.min.js	2D depiction of SMILES strings	MIT
sql.js (SQLite compiled to WASM)	lib/sql-wasm.js , lib/sql-wasm.wasm	Client-side SQL playground	MIT

The exact versions are those of the files included in the repository. Each permissive license requires preserving the original copyright notice, which must be kept in the library files or in a `THIRD_PARTY_NOTICES` file.

2. External services/APIs (not redistributed)

These services are queried over the network at runtime (their data is not part of the package); their use is subject to the respective terms.

Service	Use
PubChem PUG-REST (NIH/NCBI)	IUPAC/common name, CAS, formula, 3D SDF models, GHS/CLP pictograms

Service	Use
NIST WebBook · SDBS · MassBank · nmrshiftdb2 · SpectraBase	Links to reference experimental spectra
NASA / ESA	Astronomical images and data (Astrochemistry section)

3. Intellectual property of the proprietary code

• BioSpecInfo's application code (UI logic, the proprietary SVG mechanism engine, the CIP stereochemistry viewer in Canvas, the heuristic spectral predictors, the educational content, the internal databases) is the **author's original work**, except for the third-party libraries listed above.

• **No copyleft-licensed code** is embedded in the proprietary source: the author retains full licensing freedom over their own code.

3.1 Recommendation

The repository does not yet contain a `LICENSE` file explicitly stating the terms of the proprietary code. It is recommended to add one, choosing the desired license (e.g. **MIT** for maximum adoption, or a proprietary "all rights reserved" license to restrict reuse), together with a `THIRD_PARTY_NOTICES.md` file collecting the copyright notices of the libraries.

4. Compliance summary

Item	Result
Viral copyleft dependencies (GPL/AGPL)	None
Dependency licenses	Permissive (MIT / BSD-3-Clause)
Obligation to keep copyright notices	Yes (in library files / NOTICES)
Licensing freedom over proprietary code	Full
Project <code>LICENSE</code> file	⚠ To be added (recommended)